

Generation Interconnection System Impact Study Report

For Queue Project Q14-0362

Home 138 kV
October 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0362. The Interconnection Customer proposes a 295.3 MW wind facility interconnecting at the Home 138 kV substation in the North Carolina area. The request was received on August 13, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 215 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,190,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BELTSVILLE – BUNKER HILL 138kV	TABER – GLENANNAN 345kV Line	117.0	586	5.9
BUNKER HILL – NASA 138kV L	TABER SUBSTATION – GLENANNAN 345kV	115.0	629	5.9
JEROME – HIGH RIDGE 138kV line	TABER – DIXLETA 345kV	113.9	601	8.1
HOME – PHELOM PARTS 138kV	HOME 138/1 kV XFMR	112.9	204	9.4
TABER SUBSTATION – GLENANNAN 345kV	SIERRA PACIFIC LUMBER 138/1 kV Autotransformer	112.4	1,656	20.3
NORTH POLE – DIXLETA 138kV Ckt 1	TABER – DIXLETA 345kV line	112.4	338	5.1
YADKIN – HARBORGEN 138kV	WHEELABRATOR PORTSMOUTH – YADKIN 138kV Line	109.7	508	15.7
TABER – DIXLETA 345kV	SIERRA PACIFIC LUMBER 138/1 kV TX	109.5	1,619	18.3
SHARYLAND – BIXBY HILL 345kV	TOMBSTONE – BIXBY HILL 500kV	107.4	1,851	7.3
SIERRA PACIFIC LUMBER – LAKE CRK 138kV	TABER – GLENANNAN 345kV	107.1	612	5.5
TOMBSTONE – SHARYLAND 345kV	BIXBY HIL – TOMBSTONE 500kV	106.1	1,905	7.2
WHEELABRATOR PORTSMOUTH – YADKIN 138kV	YADKIN SUBSTATION – HARBORGEN 138kV	105.9	632	19.2
SUNBRIGHT – FAIRLAND 138kV	GLENANNAN – TABER 345kV Line	105.5	673	7.1
SIERRA PACIFIC LUMBER 138-1 kV Autotransformer	TABER – GLENANNAN 345kV	104.1	963	7.2
SIERRA PACIFIC LUMBER 345 kV/1 kV Auto	GLENANNAN – TABER 345kV	104.1	963	7.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SUNBRIGHT – COLESVILLE 138kV	TABER – GLENANNAN SUBSTATION 345kV	104.0	736	7.1
DRUM – DIXLETA 138kV	GLENANNAN SUBSTATION – TABER 345kV L	101.2	648	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BELTSVILLE – BUNKER HILL 138kV	Rebuild line with higher-capacity conductor	\$922,000	11% (\$100,000)
NASA – BUNKER HILL 138kV	Rebuild line with higher-capacity conductor	\$1,961,000	11% (\$212,000)
JEROME – HIGH RIDGE 138kV L	Rebuild line with higher-capacity conductor	\$1,295,000	11% (\$139,000)
HOME – PHELOM PARTS 138kV L	Rebuild line with higher-capacity conductor	\$8,555,000	16% (\$1,412,000)
TABER – GLENANNAN 345kV	Rebuild line with higher-capacity conductor	\$58,750,000	11% (\$6,332,000)
NORTH POLE SUBSTATION – DIXLETA 138kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,703,000	19% (\$318,000)
YADKIN – HARBORGEN 138kV	Reconductor line	\$1,271,000	12% (\$156,000)
TABER – DIXLETA 345kV	Reconductor line	\$9,488,000	11% (\$1,022,000)
SHRYLAND SUBSTATION – BIXBY HILL 345kV	Reconductor line	\$14,525,000	11% (\$1,540,000)
SIERRA PACIFIC LUMBER – LAKE CRK 138kV	Reconductor line	\$947,000	11% (\$102,000)
TOMBSTONE – SHARYLAND 345kV	Reconductor line	\$14,705,000	11% (\$1,555,000)
YADKIN – WHEELABRATOR PORTSMOUTH 138kV	Reconductor line	\$1,553,000	12% (\$193,000)
SUNBRIGHT – FAIRLAND 138kV	Reconductor line	\$204,000	11% (\$22,000)
SIERRA PACIFIC LUMBER 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$5,191,000	11% (\$561,000)
SIERRA PACIFIC LUMBER 345 kV/1 kV Auto	Replace transformer with higher-rated unit	\$12,852,000	11% (\$1,388,000)
SUNBRIGHT – COLESVILLE 138kV	Reconductor line	\$565,000	11% (\$60,000)
DIXLETA – DRUM 138kV	Reconductor line	\$737,000	11% (\$78,000)
	Total allocated to project		\$15,190,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.